SMAI Assignment-4 (Theory)

1.1.1 Softmax & Cross Entropy loss

(a) Given loss equation

$$l = -\ln p_y \quad \text{where } p_y = \frac{e^{o_y}}{\sum_{j=1}^{K} e^{o_j}}$$

$$l = -\ln\left(\frac{e^{o_y}}{\sum_{j=1}^{K} e^{o_j}}\right)$$

$$l = -\left(\ln(e^{o_y}) - \ln\left(\sum_{j=1}^{K} e^{o_j}\right)\right)$$

$$l = -\left(o_y - \ln \sum_{j=1}^{K} e^{o_j}\right)$$

$$\therefore l = \ln\left(\sum_{k=1}^{K} e^{o_k}\right) - o_y$$

Subtracting same constant c from every logit leaves softmax probabilities unchanged.

Suppose new $o'_k = o_k - c$

$$p'_k = \frac{e^{o_k - c}}{\sum_{j=1}^{K} e^{o_j - c}} = \frac{e^{o_k} e^{-c}}{\sum_{j=1}^{K} e^{o_j} e^{-c}}$$

$$= \frac{e^{o_k} \not{e^{-c}}}{\not{e^{-c}} \sum_{j=1}^{K} e^{o_j}} = \frac{e^{o_k}}{\sum_{j=1}^{K} e^{o_j}} = p_k$$

choosing $c = \max_k o_k$ prevents floating point overflow, without this shift if any logit o_k is moderately large, then evaluating e^{o_k} causes overflow. Hence subtracting c helps to keep the exponent $e^{o_k - c}$ in range $(0, 1]$.

(b) Derive gradient of CE loss wrt each logit O_K.

$$l = \ln\left(\sum_{K=1}^{K} e^{O_K}\right) - O_y$$

$$\frac{\partial l}{\partial O_K} = \frac{\partial}{\partial O_K}\left(\ln\left(\sum_{K=1}^{K} e^{O_K}\right) - O_y\right)$$

$$= \frac{\partial}{\partial O_K}\left[\ln\left(\sum_{K=1}^{K} e^{O_K}\right)\right] - \frac{\partial(O_y)}{\partial O_K}$$

$$= \frac{1}{\sum_{K=1}^{K} e^{O_K}} \cdot \frac{\partial}{\partial O_K}\left(\sum_{K=1}^{K} e^{O_K}\right) - \frac{\partial O_y}{\partial O_K}$$

$\frac{\partial}{\partial O_K}\left(\sum_{K=1}^{K} e^{O_K}\right)$ when $j=K$ its e^{O_K} else 0

$$= \frac{1}{\sum_{K=1}^{K} e^{O_K}} e^{O_K} - \frac{\partial O_y}{\partial O_K}$$

$$= \frac{e^{O_K}}{\sum_{K=1}^{K} e^{O_K}} - \frac{\partial O_y}{\partial O_K}$$

$$= P_K - \frac{\partial O_y}{\partial O_K}$$

$$\frac{\partial O_y}{\partial O_K} \begin{cases} 1 & \text{when } K = y \\ 0 & \text{when } K \neq y \end{cases}$$

$$\therefore = P_K - 1[K = y]$$

Forward and Backward pass by hand

(a) Forward Pass

at $L=1$

$x = (1, 2)$

$$z^{(1)} = xW^{(1)} + b^{(1)} = (1\ 2)\begin{pmatrix} 0.5 & -0.3 \\ 0.2 & 0.4 \end{pmatrix} + (0\ -1)$$

$$= (0.5+0.4 \quad -0.3+0.8) + (0\ -1)$$

$$= (0.9 \quad 0.5) + (0\ -1)$$

$$z^{(1)} = (0.9 \quad -0.5)$$

Apply Relu $\phi(t) = \max(0, t)$

$$h^{(1)} = \phi(z^{(1)}) = (\max(0, 0.9) \quad \max(0, -0.5)) = (0.9\ 0)$$

at $L=2$

$$O = z^{(2)} = h^{(1)} W^{(2)} + b^{(2)} = (0.9\ 0)\begin{pmatrix} 0.6 & -0.2 & 0.1 \\ -0.4 & 0.3 & 0.5 \end{pmatrix} + (0.2\ 0\ -0.1)$$

$$= (0.9\times0.6 + 0\times-0.4 \quad 0.9\times-0.2 + 0\times0.3 \quad 0.9\times0.1 + 0\times0.5) + (0.2\ 0\ -0.1)$$

$$O = (0.74 \quad -0.18 \quad -0.01)$$

Softmax and CE loss

$$\text{Softmax } P_k = \frac{e^{O_k}}{\sum_{j=0}^{2} e^{O_j}}$$

$$e^{O_0} = e^{0.74} \approx 2.095936$$

$$e^{O_1} = e^{-0.18} \approx 0.835270$$

$$e^{O_2} = e^{-0.01} \approx 0.990050$$

$$\sum_{k=0}^{2} e^{O_k} = 3.921256$$

$$P_0 = \frac{2.095936}{3.921256} \approx 0.5345$$

$$P_1 = \frac{0.835270}{3.921256} \approx 0.2130$$

$$P_2 = \frac{0.99005}{3.921256} \approx 0.2525$$

$$\therefore P = (0.5345 \quad 0.2130 \quad 0.2525)$$

∴ for true label $y=1$

$$l = -\ln P_1 = -\ln(0.2130) \approx 1.5464$$

(b) Backward Pass

$\delta^{(2)}$

onehot(1) = (0 1 0)

$\delta^{(2)}$ = P − onehot(1)

= (0.534481 0.213011 − 1 0.252483)

= (0.534481 −0.7870 0.252483)

$\delta^{(1)}$

Derivative of Relu $\phi'(t) = 1[t>0]$ evaluated at $z^{(1)} = (0.9 \quad -0.5)$

$\phi'(z^{(1)}) = (1 \quad 0)$

backprop error before activation

$$\delta^{(2)}(W^{(2)})^T = (0.534481 \quad -0.786989 \quad 0.252483)\begin{pmatrix} 0.6 & -0.4 \\ -0.2 & 0.3 \\ 0.1 & 0.5 \end{pmatrix}$$

$\delta^{(2)}(W^{(2)})^T$ = (0.534481(0.6) + 0.786989(0.2) + 0.252483(0.1) 0.534481(−0.4) − 0.786989(0.3) + 0.252483(0.5)

= (0.503335 −0.323647)

$$\delta^{(1)} = (\delta^{(2)}(W^{(2)})^T) \odot \phi'(z^{(1)})$$

$$= (0.503335 \quad -0.323647) \odot (1 \quad 0)$$

$$= (0.503335 \quad 0)$$

Param Gradients

$$\frac{\partial l}{\partial W^{(2)}} = (h^{(1)})^T \delta^{(2)} = \begin{pmatrix} 0.9 \\ 0 \end{pmatrix} (0.534481 \quad -0.786989 \quad 0.252483)$$

$$= \begin{pmatrix} 0.4810 & -0.7083 & 0.2272 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\frac{\partial l}{\partial b^{(2)}} = \delta^{(2)} = (0.534481 \quad -0.786989 \quad 0.252483)$$

$$\frac{\partial l}{\partial W^{(1)}} = (h^{(0)})^T \delta^{(1)} = x^T \delta^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} (0.503335 \quad 0)$$

$$= \begin{pmatrix} 0.503335 & 0 \\ 1.0067 & 0 \end{pmatrix}$$

$$\frac{\partial l}{\partial b^{(1)}} = \delta^{(1)} = (0.503335 \quad 0)$$

The second layer of hidden layer $h_2^{(1)}$ receives a zero-gradient for both $w_2^{(1)}$ and $b_2^{(1)}$. Its pre-activation input $z_2^{(1)} = -0.5$ is negative. Because Relu is flat for negative inputs, the local derivative is 0 blocking gradient flow.

(c) Gradient descent update

$\eta = 0.1$

layer 1

$$W^{(1)} \leftarrow W^{(1)} - 0.1 \cdot \frac{\partial l}{\partial W^{(1)}} = \begin{pmatrix} 0.5 & -0.3 \\ 0.2 & 0.4 \end{pmatrix} - 0.1 \begin{pmatrix} 0.503335 & 0 \\ 1.00662 & 0 \end{pmatrix} = \begin{pmatrix} 0.4497 & -0.3 \\ 0.0993 & 0.4 \end{pmatrix}$$

$$b^{(1)} \leftarrow b^{(1)} - 0.1 \frac{\partial l}{\partial b^{(1)}} = (0 \quad -1) - 0.1(0.503335 \quad 0) = (-0.0503 \quad -1)$$

Layer 2

$$W^{(2)} \leftarrow W^{(2)} - 0.1 \frac{\partial l}{\partial W^{(2)}} = \begin{pmatrix} 0.6 & -0.2 & 0.1 \\ -0.4 & 0.3 & 0.5 \end{pmatrix} - 0.1 \begin{pmatrix} 0.481033 & -0.70829 & 0.227235 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0.5519 & -0.1292 & 0.0773 \\ -0.4 & 0.3 & 0.5 \end{pmatrix}$$

$$b^{(2)} \leftarrow b^{(2)} - 0.1 \cdot \frac{\partial l}{\partial b^{(2)}} = (0.2 \quad 0 \quad -0.1) - 0.1(0.536481 \quad -0.786989 \quad 0.252485)$$

$$= (0.1466 \quad 0.0787 \quad -0.1252)$$

Recompute loss

$$z^{(1)}_{new} = (1 \quad 2)\begin{pmatrix} 0.46966 & -0.3 \\ 0.09433 & 0.4 \end{pmatrix} + (-0.050334 \quad -1)$$

$$= (0.46966 + 2(0.09933) \quad -0.3 + 2(0.4)) + (-0.050334 \quad -1)$$

$$= (0.598 \quad -0.5)$$

$$h^{(1)}_{new} = \phi(z^{(1)}_{new}) = (0.598 \quad 0)$$

$$O_{new} = (0.598 \quad 0)\begin{pmatrix} 0.551897 & -0.129171 & 0.077237 \\ -0.4 & 0.3 & 0.5 \end{pmatrix} + (0.146552 \quad 0.078699 \quad -0.125248)$$

$$= (0.598(0.551897) \quad 0.598(-0.129171) \quad 0.598(0.77237)) + (0.146552 \quad 0.078699 \quad -0.125248)$$

$$= (0.4766 \quad 0.0015 \quad -0.0790)$$

$$e^{o_{0,new}} = e^{0.4766} \approx 1.610557 \qquad \sum_{k=0}^{2} e^{o_{k,new}} = 1.610557 + 1.001456 + 0.92399$$

$$e^{o_{1,new}} = e^{0.001455} \approx 1.001456 \qquad = 3.536012$$

$$e^{o_{2,new}} = e^{-0.079032} \approx 0.92399 \qquad P_{1,new} = \frac{1.001456}{3.536012} \approx 0.2832$$

$$l_{new} = -\ln P_{1,new} = -\ln(0.2832) \approx 1.2615$$

∴ Yes loss decreased from 1.5464 to 1.2615

1.3 MLP vs IOC-NN

1.3.1 (a)

Property	MLP	IOC-NN
Hidden-layer activation	ReLU $\max(0, z)$	ELU$(z) \begin{cases} z & z > 0, \\ e^z - 1 & z \le 0 \end{cases}$
First layer weights $W^{(1)}$	Unconstrained Sign	Unconstrained Sign
Weights $W^{(2)}, W^{(3)}, \ldots$	Any Sign	Non-negatives real values
Biases	Any Sign	Any Sign
Initialisation of $W^{(2)}, W^{(3)}, \ldots$	Normal, Variance $2/n_{in}$ for ReLU, else $1/n_{in}$	Uniform on $(0, 2/n_{in})$
What happens after each gradient step?	Standard update; no projection	Replace negative weights with e^{w-5}

Properties which are same are,

1. First layer weights $W^{(1)}$
2. Biases
3. The overall architecture, loss and gradient-descent procedure are all same.

(b) The network is called input-output convex because each output logit before softmax is a convex function of the input vector x. when evaluating the convexity property, all the network parameters including all weight matrices $W^{(l)}$ and bias vectors $b^{(l)}$ are kept fixed. the total function is convex in x because we use convex, non decreasing activation like ELU along with non-negative weights in all layers therefore preserving convexity.